Signal State: GREEN (Score: 3552)

PRACTICAL NO 2 : MULTI-CLASS CLASSIFICATION APPLICATION FOR DETECTING DISEASED PLANTS BASED ON IMAGES OF THEIR LEAVES

```
import os
import cv2
import numpy as np
from sklearn.neighbors import KNeighborsClassifier

x=[]
y=[]

dataset ="PlantVillage"

#load image
for folder in os.listdir(dataset):
    path=os.path.join(dataset,folder)

    for img_name in os.listdir(path):
        img_path=os.path.join(path,img_name)

        img=cv2.imread(img_path)
        img=cv2.resize(img,(50,50))

        x.append(img.flatten()))
        y.append(folder)

#Train Model
model =KNeighborsClassifier(n_neighbors=3)
model.fit(x,y)

print("Model Trained Succesfully.")
```

```
#test img
test_img=cv2.imread("test.jpg")
test_img=cv2.resize(test_img,(50,50))
test_img=test_img.flatten().reshape(1,-1)

prediction= model.predict(test_img)

print("Predicted Disease:",prediction[0])
```

Output :



Model Trained Succesfully.
Predicted Disease: Tomato_Bacterial_spot